Organization of Gene
“Genome Replication ”

SGL - 4

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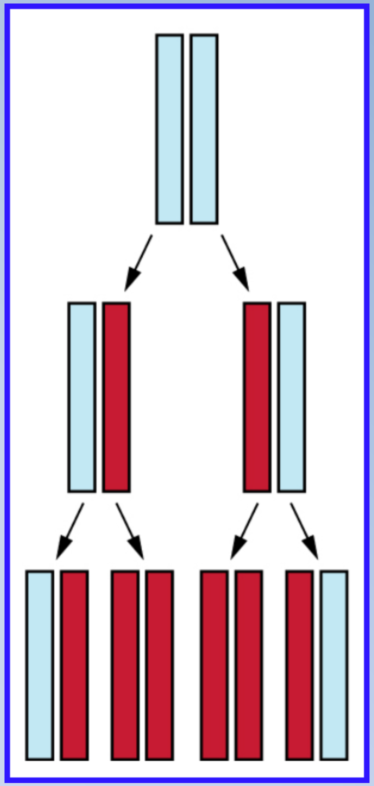
Molecular Biology :

DNA Replication: Semiconservative DNA replication

- DNA synthesis involves a highly coordinated action of many proteins. *Precision and speed* are required.
- Mainly the principal enzymatic proteins are *Polymerases*, which carry out template-directed synthesis and *Helicases*, separate the two strands to generate the replication fork.



- DNA replication is genetically relatively simple. During replication, each strand of DNA serves as a template for the formation of a new strand
- During the process of DNA replication, the two strands of the double helix separate, and each strand serves as a template from which the new complementary strand is copied.
- After replication, each double-stranded DNA includes one parental or “old” strand and one “new” strand, where called DNA semiconservative replication model

Molecular Biology :

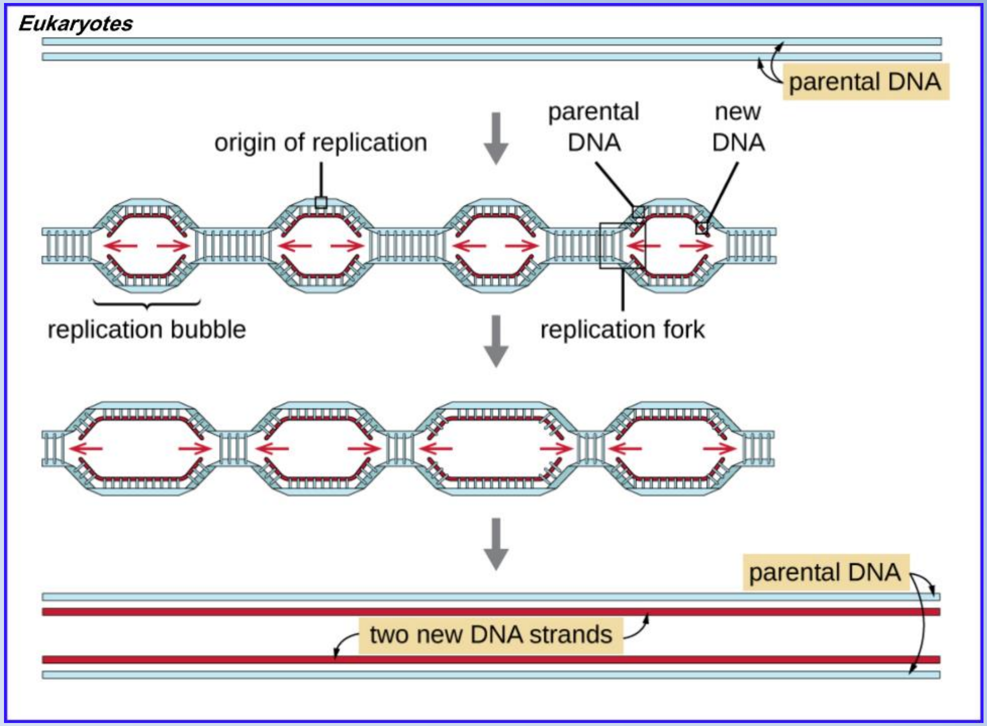
DNA Replication: Definitions _ Origin of replication

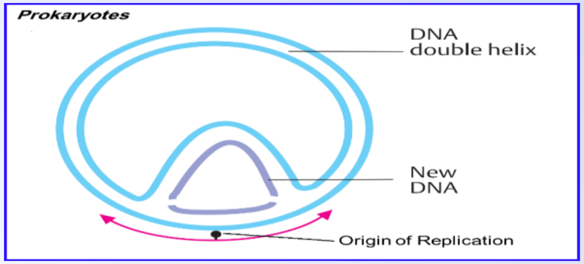
- The DNA replication occurs at specific nucleotide sequence called the *Origin of replication*.

- Most Prokaryotes has a *Single Origin of Replication* on its one chromosome. The Eukaryotes has *Multiple Origin of Replication*.

- The origin of replication is approximately 245 bp long and is rich in adenine-thymine (*A-T*) sequences.

- An enzyme called *Helicase* then separates the DNA strands by breaking the hydrogen bonds between the nitrogenous base pairs. A-T sequences have fewer hydrogen bonds and, hence, have weaker interactions than G-C





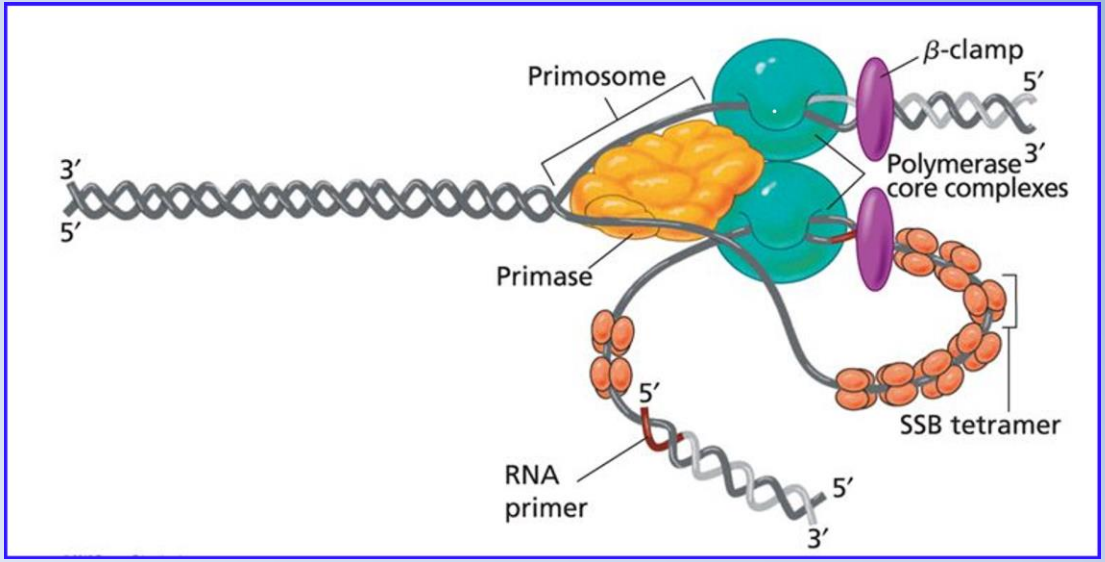
Molecular Biology :

DNA Replication: Definitions _ Replisome

The entire complex will be formed and composed of:

- *Primases* which initiate chain synthesis at preferred sites;
- *Initiation Proteins*, which recognize the origin of replication point; and
- *Proteins that remodel the double helix* are called the:

Replisome Complex



Molecular Biology :

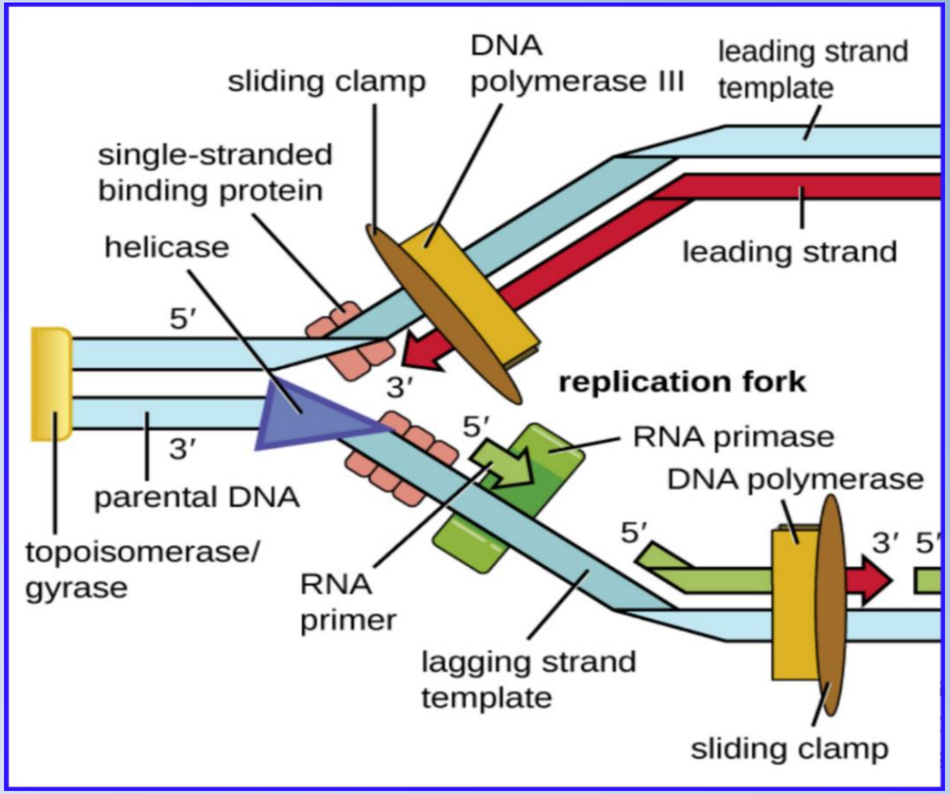
DNA Replication: Stage1 - Initiation

- DNA is typically wrapped around histones, supercoiled, extensively wrapped and twisted on it self and this packaging makes the information in the DNA molecule inaccessible.

- Some of the proteins within replisome that bonded to the origin of replication are start making single-stranded regions of DNA accessible for replication

- The initiation of DNA replication occurs at Origin of replication and replisome complex bind to the Origin of replication begin the replication process.

- The *Topoisomerases* start to change the shape and relaxing the supercoiling of the chromosome.



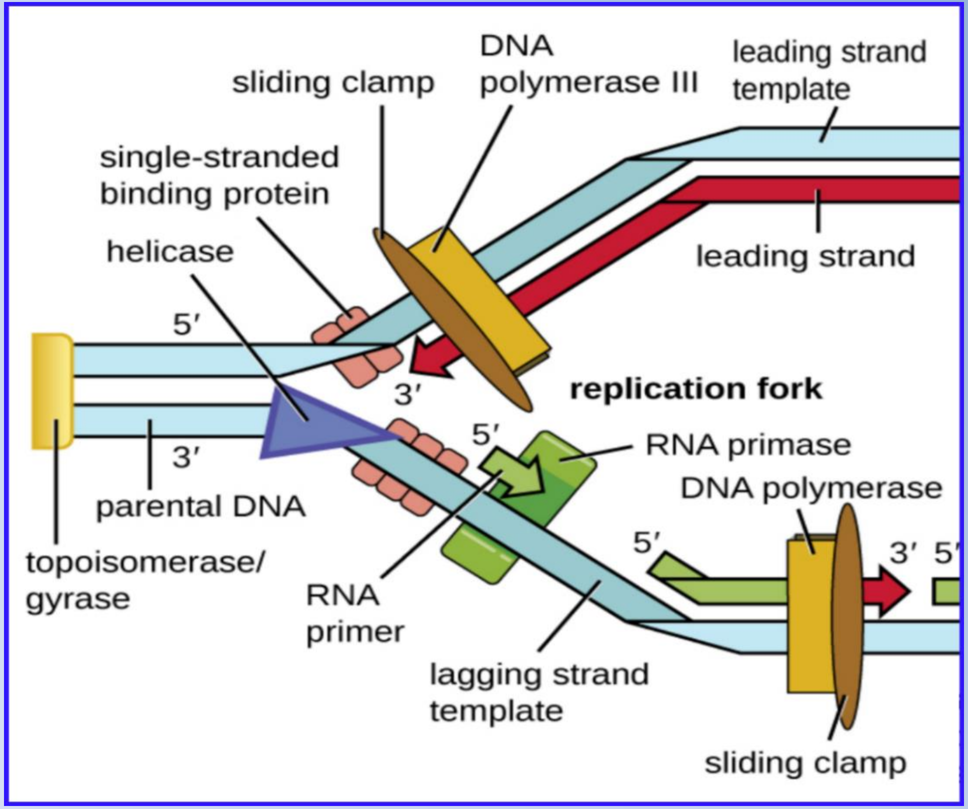
Molecular Biology :

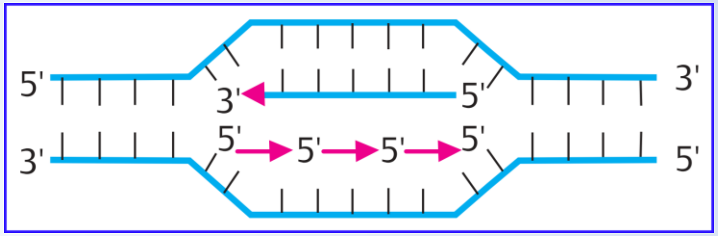
DNA Replication: Stage1 - Initiation

- The enzyme *Helicase* separates the DNA strands by breaking the hydrogen bonds between the nitrogenous base pairs, the AT sequences have fewer hydrogen bonds and have weaker interactions than (GC) sequences.

- Y-shaped structures called *Replication Forks* are formed. Two replication forks are formed at the origin of replication called replication bubble, allowing for *Bidirectional Replication*.

- The DNA near each replication fork is coated with *Binding Proteins* that prevent the single-stranded DNA from rewinding into a double helix.

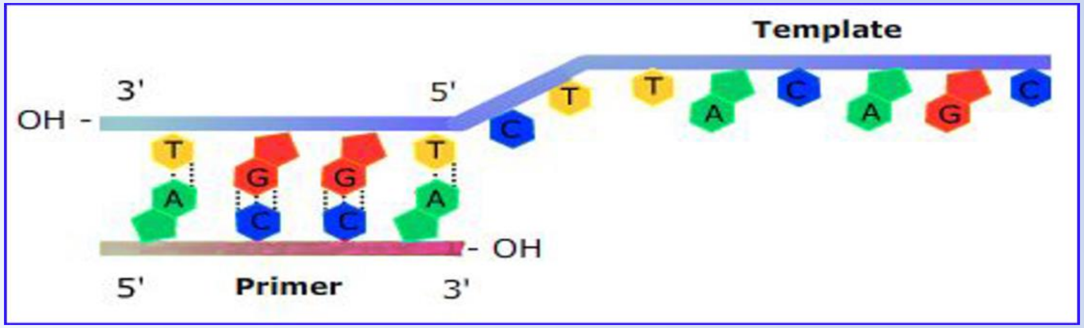


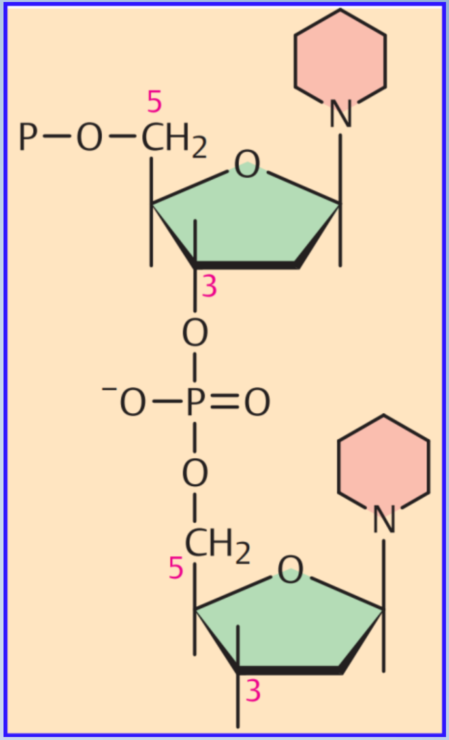


Molecular Biology :

DNA Replication: Definitions _ Primers

- DNA polymerase is able only to add nucleotides only in the 5’ to 3’ direction (a new DNA strand can be only extended in this direction).
- DNA polymerase cannot add nucleotides if a free 3’-OH group is not available, which is the case for a single strand of DNA.
- An RNA sequence called the *Primer* provide the free 3’-OH end it allows the DNA polymerase start of DNA synthesis.
- The primer is 5 to 10 nucleotides long and complementary to the parental or template DNA.
- The RNA primer is synthesized by *RNA primase*, unlike *DNA polymerases*, *RNA polymerases* do not need a free 3’-OH group to synthesize an RNA molecule.

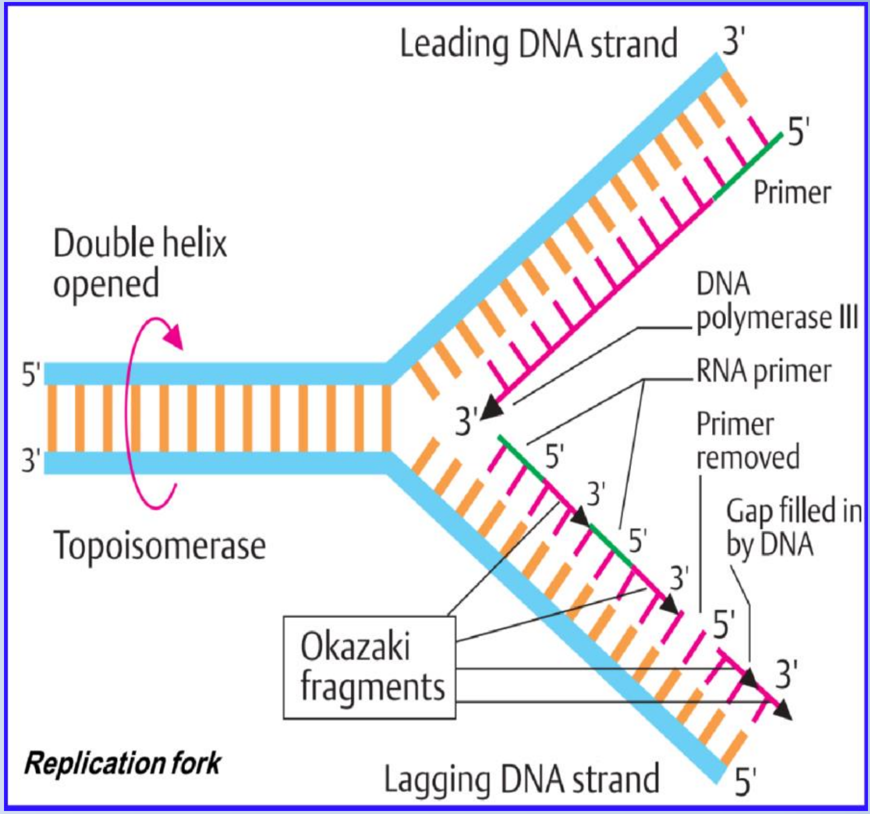


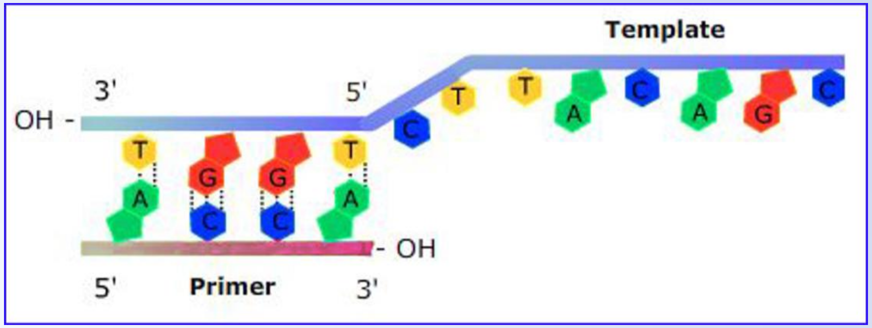


Molecular Biology :

DNA Replication: Stage 2 - Elongation

- DNA replication can begin Once single-stranded DNA is accessible at the origin of replication.
- DNA polymerase can only extend in the 5’ to 3’ direction, which poses a problem at the replication fork.
- The DNA double helix is antiparallel; that is, one strand is oriented in the 5’ to 3’ direction and the other is oriented in the 3’ to 5’ direction.
- During replication, one strand, which is complementary to the 3’ to 5’ parental DNA strand, is synthesized continuously toward the replication fork because polymerase can add nucleotides in this direction.
- This continuously synthesized strand is known as the *Leading Strand*.





Molecular Biology :

DNA Replication: Stage 2 - Elongation

- The other strand, complementary to the 5’ to 3’ parental DNA, grows away from the replication fork, where the RNA polymerase add a primer at the beginning of replication fork

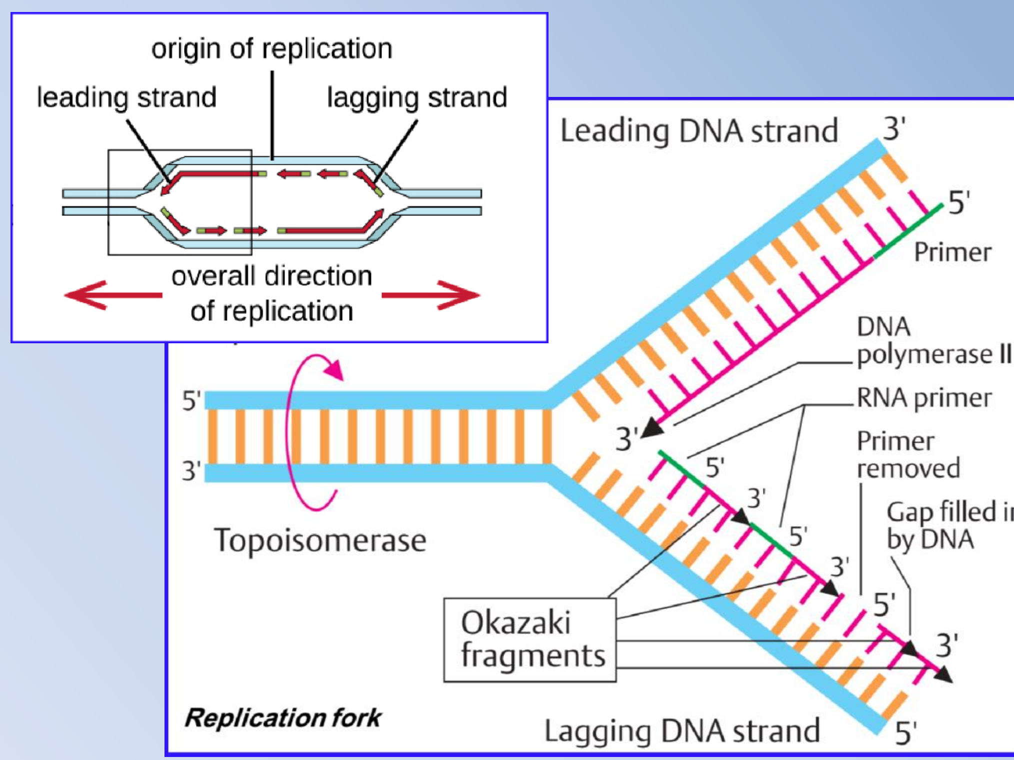
- So the polymerase must move back toward the replication fork to begin adding bases to a new primer again in the direction away from the replication fork.

- It does so until it bumps into the previously synthesized strand and then it moves back again.

- These steps produce small DNA sequence fragments known as *Okazaki* fragments and each separated by RNA primer.

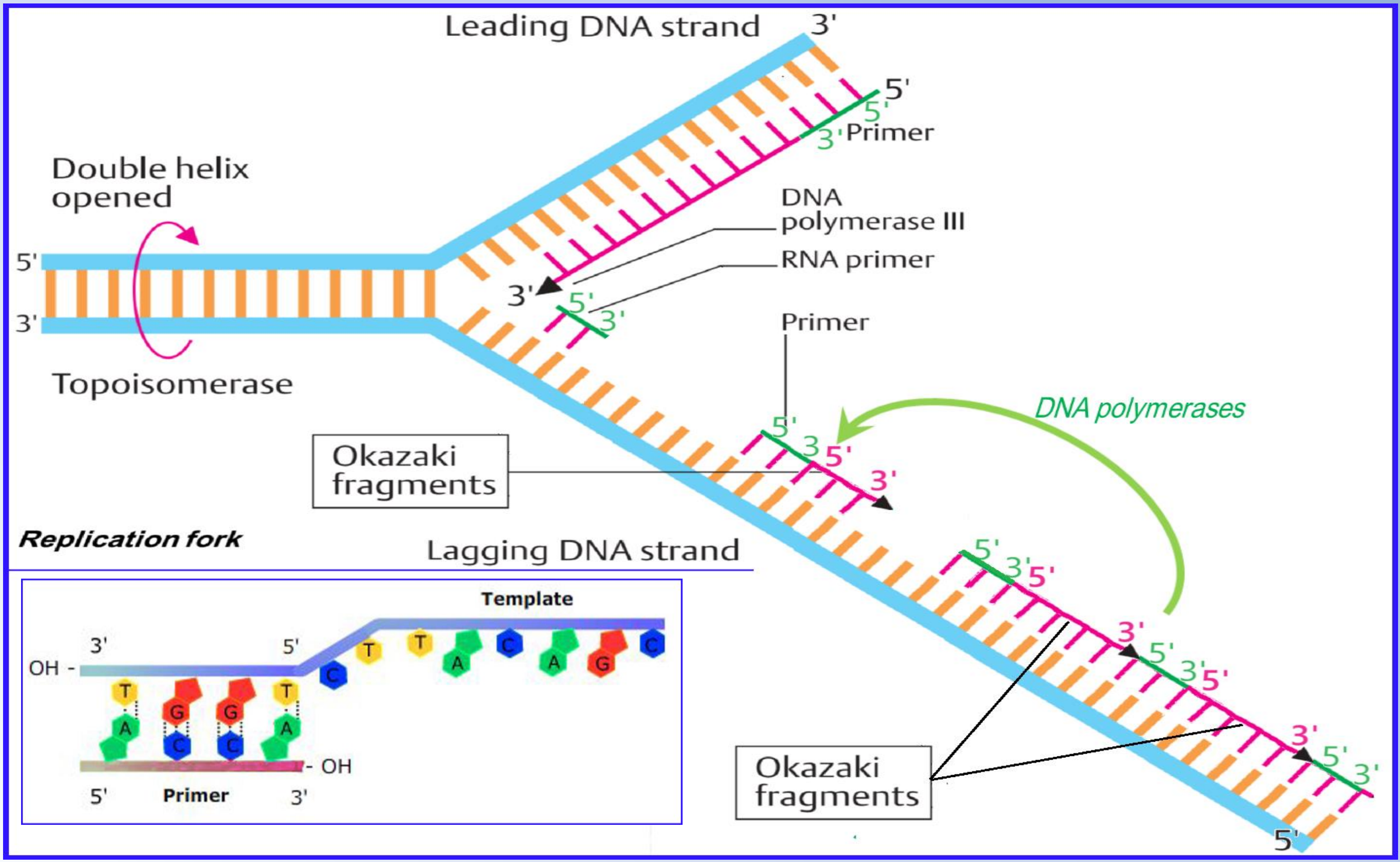
- Okazaki fragments are joined by DNA ligase & primers removed.

- The strand with the Okazaki fragments is known as the lagging strand, and its synthesis is said to be discontinuous.



Molecular Biology :

DNA Replication: Stage 2 - Elongation



Molecular Biology :

DNA Replication: Stage 3 - Termination

- Once the complete chromosome has been replicated, termination of DNA replication must occur.

- In prokaryotes, the resulting complete circular genomes of are *Concatenated*, meaning that the circular DNA chromosomes are interlocked and must be separated from each other by activity of topoisomerase.

- Eukaryotic chromosomes are linear, The ends of the linear chromosomes are known as *Telomeres* contain no genes and have a special structure.

- The telomeres (In Human) consist of noncoding repetitive sequences of six base-pair sequence, TTAGGG, is repeated 100 to 1000 times attaches to the end of the chromosome.

- The telomeres protect coding sequences from being lost as cells continue to divide.